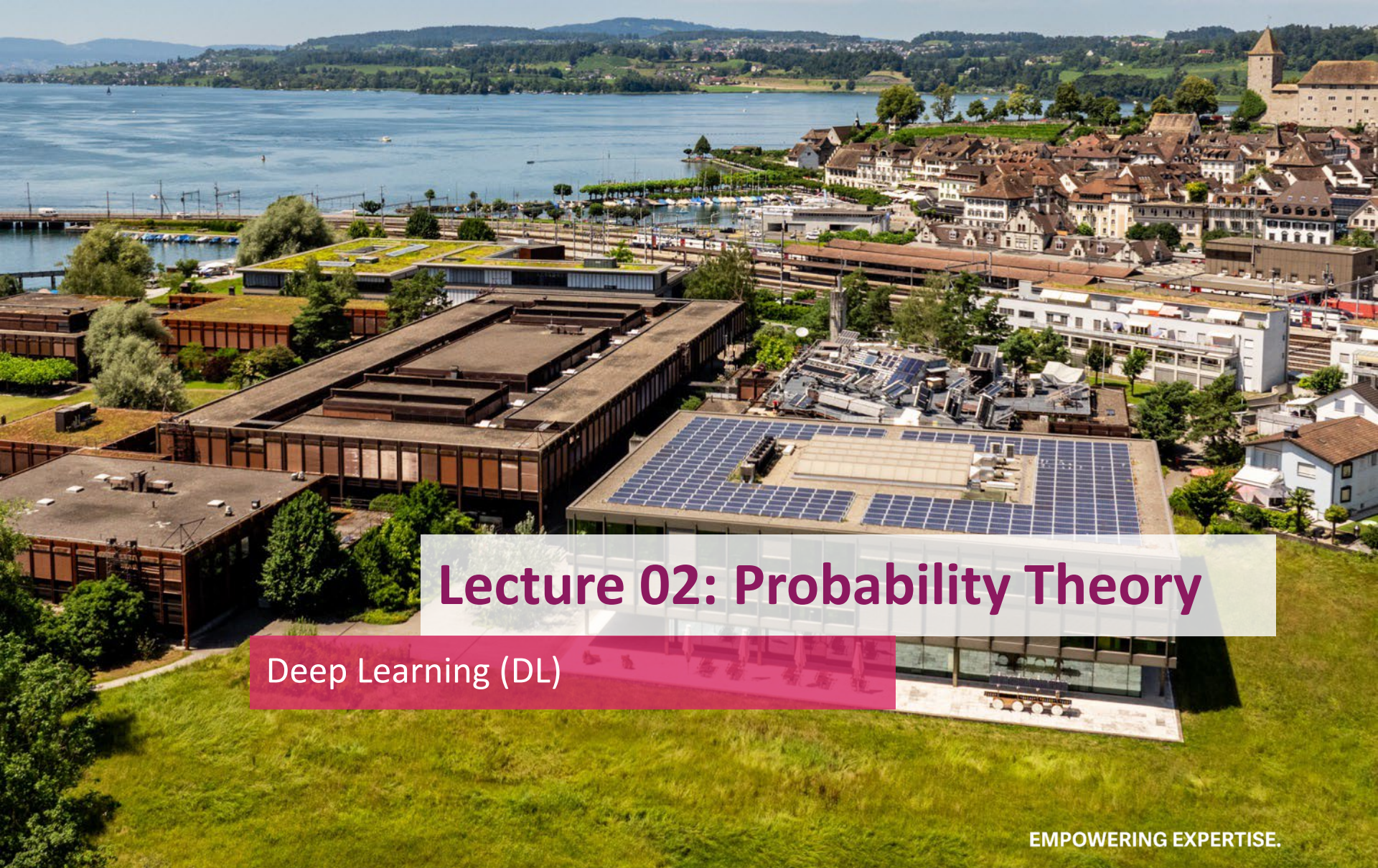




Lecture 02: Probability Theory

Deep Learning (DL)

EMPOWERING EXPERTISE.

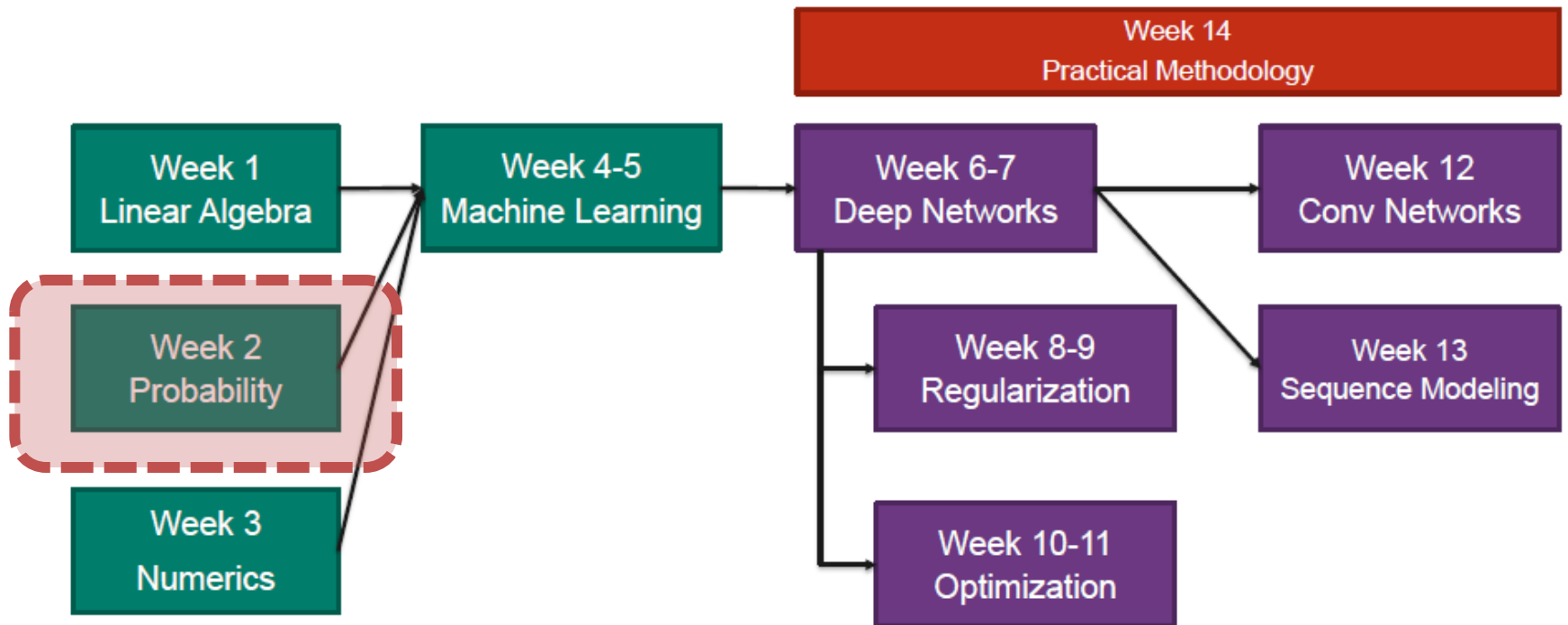


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23 February 2026



Overview of DL-Lecture



Overview of this lecture

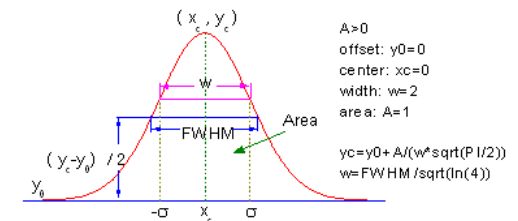
- Basics of probability
 - Probability distributions
 - Important functions
 - Information theory

Random Variables

- A **random variable** (rv) is a variable that can take on **different values** randomly
 - This does not mean, that every value is equally likely!
- There are two types of rv

Type of rv	Values of rv	Description of rv with ...
Continuous	Real numbers	Probability density function (pdf)
Discrete	Discrete set	Probability mass function (pmf)

- Example: **Random Noise** is a continuous rv
 - Often (but not always) noise follows a normal distribution



Discrete random variables

- **Discrete rv's** have a **probability mass function (PMF)**

$$x \sim P(x)$$

- Notation: $P(x)$ or $P_x(x)$
- $P(x)$ and $P(y)$ are NOT the same
- A PMF is directly the probability of x

- **Properties** of a PMF

1. Domain of $P(x)$: all possible states of x
2. $0 \leq P(x) \leq 1$
3. $\sum P(x) = 1$

- **Note:**

- $P(x) = 0$ does not mean impossible – only highly improbable
- $P(x) = 1$ does not mean guaranteed – only highly probable

Discrete random variables

- Example: **discrete uniform distribution**
 - Random variable x with k different states x_i
 - Each state is **equally likely**
 - Resulting PMF:

$$P(x = x_i) = \frac{1}{k} !$$

- **Properties**

1. Domain = all possible states
2. $0 \leq P(x) \leq 1$
3. $\sum P(x) = k \frac{1}{k} = 1$



- Examples

- Coin $\rightarrow k=2$
- Dice $\rightarrow k=6$

Continuous random variables

- **Difficulty:** can take on an **infinite number of values**

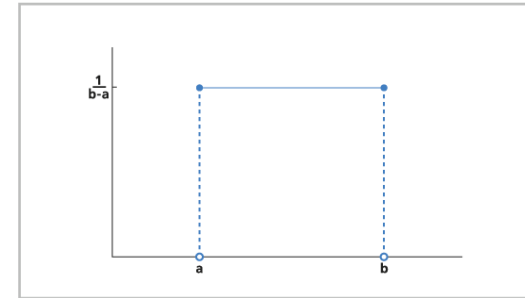
- ➔ a PMF would always be 0

- Instead of a PMF, a **probability density function (PDF)** is used

- PFM: probability

- PDF: **probability per area = likelihood**

- **Notation:** lowercase $p(x)$ or $p_x(x)$



$$x \sim p(x)$$

Notice lower case p for continuous values

- **Properties**

1. Domain of $p(x)$: all possible values of x
2. $p(x) \geq 0$ (but not ≤ 1)
3. $\int p(x) dx = 1$

Continuous random variables

- **Probabilities** only make sense **over an interval**

- Because likelihood = probability / area, a probability is obtained by integrating

$$P(a \leq x \leq b) = \int_{[a,b]} p(x) dx$$

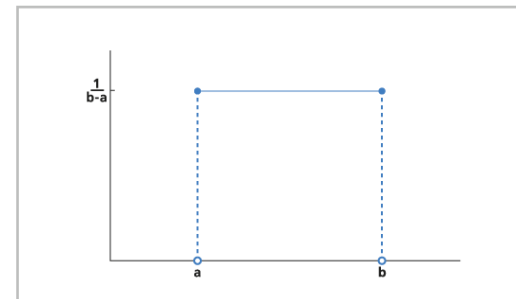
!

- Example: **continuous uniform distribution**

- Each value in the interval $[a, b]$ is equally likely
- Notation

$$x \sim U(a, b)$$

$$u(x; a, b) = \frac{1}{b-a}$$



Joint and Marginal Probability

- Joint PDF or PMF
 - 2-D (or multi-dimensional) PDF or PMF over multiple variables
- Marginal probability
 - “**Marginalization**” = obtain PMF or PDF of a single random variable from a joint PMF or PDF
 - Marginal probability = same thing as PMF or PDF of that variable

$$P(\mathbf{x}, \mathbf{y})$$

$$p(x, y)$$

$$P(\mathbf{x} = x) = \sum_y P(\mathbf{x} = x, \mathbf{y} = y)$$

$$p(x) = \int p(x, y) dy$$

Conditional Probability

- **Goal:** describe probability of one random variable, given one knows another random variable

$$P(y | x)$$

- **Examples**

- $P(\text{sales} | \text{advertisement budget})$
- $P(\text{cat} | \text{pixels of an image})$

- **Definition**

$$P(y = y | x = x) = \frac{P(y = y, x = x)}{P(x = x)}$$

Chain Rule of Conditional Probabilities

- The joint probability over multiple variables can be decomposed into a chain of conditional probabilities

$$P(x^{(1)}, \dots, x^{(n)}) = P(x^{(1)}) \prod_{i=2}^n P(x^{(i)} | x^{(1)}, \dots, x^{(i-1)})$$

- This is obtained by applying the definition shown on the right multiple times

$$P(y = y | x = x) = \frac{P(y = y, x = x)}{P(x = x)}$$

- Easier example: 3 rv's

- Step 1: treat b,c together:

$$P(a, b, c) = P(a | b, c) P(b, c)$$

- Step 2: calculate $P(b, c)$

$$P(b, c) = P(b | c) P(c)$$

- Combine everything!

$$P(a, b, c) = P(a | b, c) P(b | c) P(c)$$

Bayes' Rule

- Goal: calculate $P(x/y)$ from $P(y/x)$
 - Very common problem

$$P(x | y) = \frac{P(x)P(y | x)}{P(y)} !$$

- Example: Medical test
 - We are interested in $P(\text{sick} | \text{test})$
 - $P(\text{test} | \text{sick})$ is easy to obtain
 - $P(\text{sick})$ can also easily be obtained

$$P(\text{sick} | \text{test}) = \frac{P(\text{test} | \text{sick}) P(\text{sick})}{P(\text{test})}$$

- **$P(y)$** is often unknown, but can easily be calculated with the rule of **total probability**

$$P(y) = \sum_x P(y | x)P(x) !$$

Statistical independence

- Two random variables x and y are called **statistically independent**, if:

- Joint probability = product of marginal probabilities

$$p(x = x, y = y) = p(x = x)p(y = y) \quad !$$

- This results in the **conditional probabilities**

- Hence, knowing x **helps nothing for describing y** – and vice versa

$$p(x|y) = \frac{p(x, y)}{p(y)} = \frac{p(x)p(y)}{p(y)} = p(x)$$

$$p(y|x) = \frac{p(x, y)}{p(x)} = \frac{p(x)p(y)}{p(x)} = p(y)$$

- Short-hand notation

$$x \perp y$$

Conditional Independence

- Two random variables x and y are **conditionally independent** given a third random variable z , if

- If z is known, then x and y are independent.
- If z is unknown, they are not independent!

$$p(x = x, y = y \mid z = z) = p(x = x \mid z = z)p(y = y \mid z = z)$$

!

- Short-hand notation

$$x \perp y \mid z$$

- Application** in ML: **Naïve Bayes**
 - Features (inputs x) are considered statistically independent, given the true class is known

Expectation

- Expectation = **weighted average** over $P(x)$

- Discrete case: **sum**

$$\mathbb{E}_{x \sim P}[f(x)] = \sum_x P(x) f(x) \quad !$$

- Continuous case: **integral**

$$\mathbb{E}_{x \sim p}[f(x)] = \int p(x) f(x) dx \quad !$$

- The full notation is not used often, instead the following short-hand notations are used

- Obviously, x follows some pdf p
- If there is only one random variable, clearly the expectation is over that.

$$\mathbb{E}_x[f(x)]$$

$$\mathbb{E}[f(x)]$$

Variance

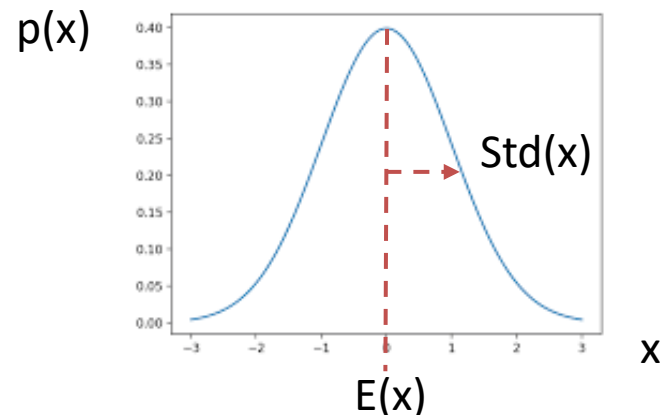
- The **variance** σ^2 measures the variability around the expectation
 - Low variance $\rightarrow$ all values x will be close around the expectation $E(x)$
 - High variance $\rightarrow$ values x can vary much from the expectation $E(x)$
- The **standard deviation** σ is the square root of the variance
 - Advantage: same unit as $f(x)$
- Example: Gaussian PDF

$$\text{Var}(f(x)) = \mathbb{E} \left[(f(x) - \mathbb{E}[f(x)])^2 \right] !$$

Squared $\rightarrow$ Square Dollars e.g. ? $\rightarrow$ Difficult to interpret

$$\text{Std}(f(x)) = \sqrt{\text{Var}(f(x))} !$$

Easy to interpret



Covariance and Correlation

- Covariance σ_{xy} Measure of **linear relation** between **two random variables** x and y

- **Sign:** positive or negative relationship
- **Absolute value:** scale of relationship and of individual variability
- Note: outer expectation is over both x and y , hence over their joint distribution $p(x, y)$

!

$$\text{Cov}(f(x), g(y)) = \mathbb{E}[(f(x) - \mathbb{E}[f(x)])(g(y) - \mathbb{E}[g(y)])]$$

$$E_{x,y \sim p(x,y)}[\dots]$$

- **Correlation:** covariance divided by the standard deviations

- Range: $[-1, +1]$
- ➔ independent of individual variabilities of x and y

$$\text{Corr}(f(x), g(y)) = \frac{\text{Cov}(f(x), g(y))}{\text{Std}(f(x))\text{Std}(g(Y))}$$
 !

Covariance matrix

- **Extension** of variance and covariance **to random vectors**
- **Definition** of covariance matrix

$$\text{Cov}(\mathbf{x})_{i,j} = \text{Cov}(x_i, x_j) \quad !$$

- **Example:** covariance matrix of a 3-d random vector

$$\text{Cov}(\mathbf{x}) = \begin{bmatrix} \text{Cov}(x_1, x_1) & \text{Cov}(x_1, x_2) & \text{Cov}(x_1, x_3) \\ \text{Cov}(x_2, x_1) & \text{Cov}(x_2, x_2) & \text{Cov}(x_2, x_3) \\ \text{Cov}(x_3, x_1) & \text{Cov}(x_3, x_2) & \text{Cov}(x_3, x_3) \end{bmatrix}$$

- The **covariance matrix** is **symmetric** because

$$\text{Cov}(x_i, x_j) = \text{Cov}(x_j, x_i)$$

- The diagonal contains the variances

$$\text{Cov}(x_i, x_i) = \text{Var}(x_i)$$

Overview of this lecture

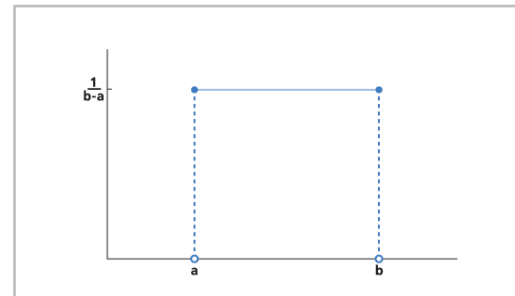
- Basics of probability
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Uniform Distribution

- Repetition from before
 - In general: **all possible values** are **equally likely**
- Discrete uniform distribution
 - k possible **states** are **equally likely**
- Continuous uniform distribution
 - All **values in interval $[a, b]$** are possible and **equally likely**

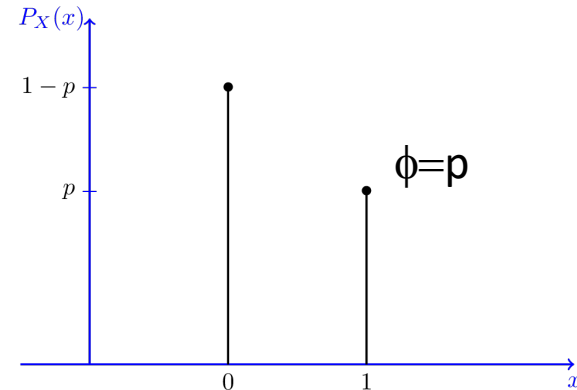
$$P(x = x_i) = \frac{1}{k}$$

$$p(x) = \frac{1}{b - a}$$



Bernoulli Distribution

- Discrete random variable with **two states**: 0 and 1
 - ➔ for **binary** random variables
- Single parameter ϕ or p , which is defined as:
 - Because only two states, $P(x = 0)$ can be calculated as
 - Both can be elegantly combined into one equation
- Properties of the Bernoulli distribution



$$P(x = 1) = \phi$$

!

$$P(x = 0) = 1 - \phi$$

!

$$P(x = x) = \phi^x (1 - \phi)^{1-x}$$

$$\mathbb{E}_x[x] = \phi$$

$$\text{Var}_x(x) = \phi(1 - \phi)$$

Multinoulli Distribution

- Extension of Bernoulli to **discrete variables** with **k different states**
- Parameter vector $\mathbf{p}$ with $k-1$ probabilities where
- **Note:** for k states, only $k-1$ parameters are required, as they must sum to 1
 - The final probability can be calculated with
 - For Bernoulli with 2 states, only 1 parameter was required
- **Application:** describe probabilities of multiple classes

$$\mathbf{p} \in [0, 1]^{k-1}$$

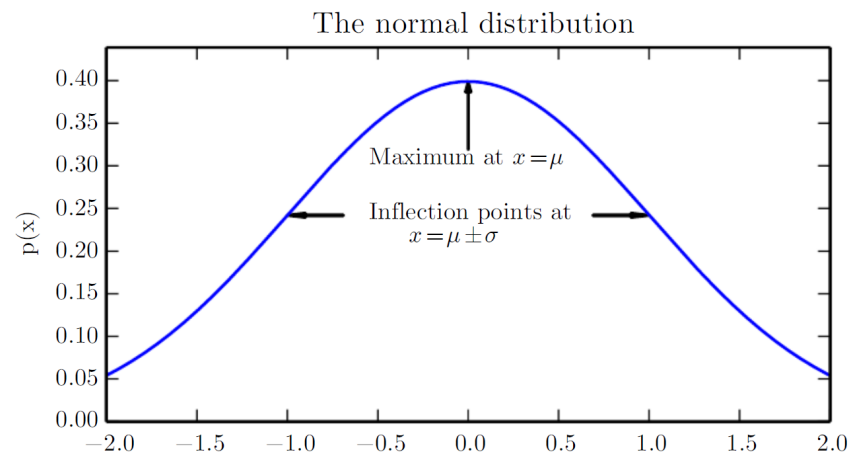
$$p_i = P(x = x_i)$$

$$p_k = 1 - \sum_{i=0}^{k-1} p_i = 1 - \mathbf{1}^T \mathbf{p}$$

Gaussian Distribution

- Very common variable for uniform distributions: **Gaussian**
 - Also called “**normal**” distribution, hence the notation $\mathcal{N}$
- Two parameters
 - μ the **mean** (expected value) and
 - σ the **standard deviation** (σ^2 the variance)

$$\mathcal{N}(x; \mu, \sigma^2) = \sqrt{\frac{1}{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x - \mu)^2\right) !$$



DL, Fig. 3.1

Gaussian Distribution

- **Alternative parameterization**
 - β is called the **precision**
 - Note: β is simply $1/\sigma^2 \rightarrow$ still the same distribution
 - Advantage: no inverse required $\rightarrow$ more computationally efficient, especially in multi-dimensional cases
- Normal distributions are a popular because of
 - 1) The **central limit theorem**, which states that the **sum of many independent rvs** approaches a **Gaussian** distribution
 - 2) For a **given variance**, the Gaussian distribution has the **largest uncertainty** (differential entropy)

$$\mathcal{N}(x; \mu, \sigma^2) = \sqrt{\frac{1}{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x - \mu)^2\right)$$

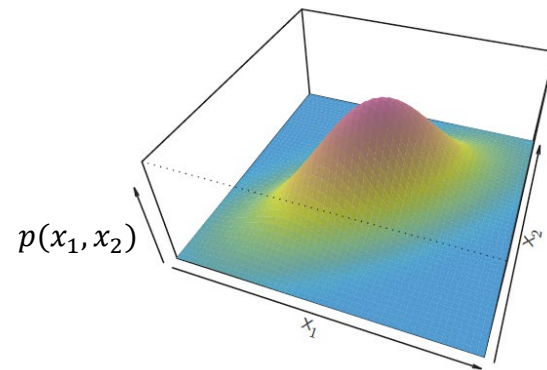
$$\mathcal{N}(x; \mu, \beta^{-1}) = \sqrt{\frac{\beta}{2\pi}} \exp\left(-\frac{1}{2}\beta(x - \mu)^2\right)$$

Gaussian Distribution

- Multidimensional Gaussian
 - These concepts can be extended to **n-dimensional random vectors**

$$\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \sqrt{\frac{1}{(2\pi)^n \det(\boldsymbol{\Sigma})}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)!$$

- Parameters
 - Mean vector $\boldsymbol{\mu}$
 - Covariance matrix $\boldsymbol{\Sigma}$



ISLP, Fig. 4.5

Gaussian Distribution

- Alternative parametrization

- The **precision matrix** β can be used instead of the covariance matrix Σ
- Advantage: **no matrix inverse** required

$$\mathcal{N}(x; \mu, \Sigma) = \sqrt{\frac{1}{(2\pi)^n \det(\Sigma)}} \exp \left(-\frac{1}{2} (x - \mu)^\top \Sigma^{-1} (x - \mu) \right)$$

$$\mathcal{N}(x; \mu, \beta^{-1}) = \sqrt{\frac{\det(\beta)}{(2\pi)^n}} \exp \left(-\frac{1}{2} (x - \mu)^\top \beta (x - \mu) \right)$$

- For **high-dimensional vectors**, the **covariance matrix** can become very **large**

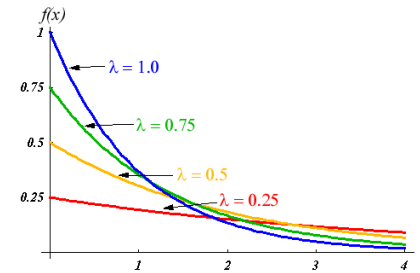
- Sometimes, the covariance matrix is **constrained** to be **diagonal**
- **Assumption**: dimensions of x uncorrelated

- Isotropic Gaussian

- “isotropic” = same in all directions
- Same variance for all dimensions, no covariance

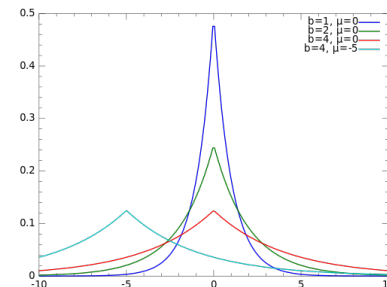
Exponential and Laplace Distribution

- **Idea:** 0 is more likely than non-zero values
- **Exponential distribution**
 - Function $\mathbf{1}_{x \geq 0}()$ is a step function: 0 for negative, 1 for positive inputs
 - Parameter λ controls steepness and hence height of peak at 0
- **Laplace distribution**
 - Related to Gaussian, but with absolute value instead of square
 - Peak at mean μ
 - Parameter γ controls steepness (and variance)



$$p(x; \lambda) = \lambda \mathbf{1}_{x \geq 0} \exp(-\lambda x) \quad !$$

$$\text{Laplace}(x; \mu, \gamma) = \frac{1}{2\gamma} \exp\left(-\frac{|x - \mu|}{\gamma}\right) \quad !$$

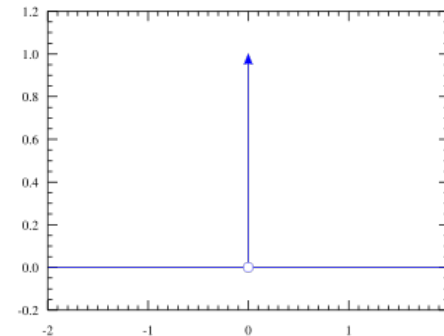


Dirac Distribution

- **Dirac function / dirac impulse**
 - Infinitely high value in an infinitely small range
 - Interesting fact: area under dirac function (integral) = 1, hence
- The function is 0 everywhere, except at 0, there it goes to infinity
 - But integrating over 0 results in 1
 - Hence, this allows the replacement of pmf by pdf, simply put a Dirac delta function at all the locations where the pmf has probability
 - For example, the way this pdf $p(x)$ is defined, the probability mass of 1 is all at μ , hence the pdf is zero everywhere, except at μ , where it is infinite. But the probability at μ is 1, since it is the integral over an infinitesimal interval around 0

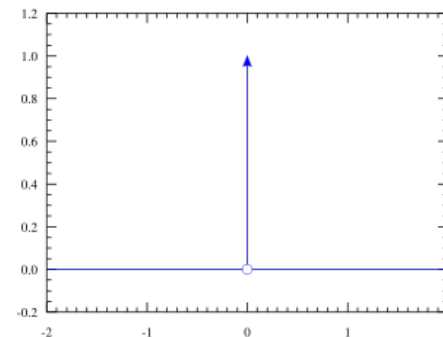
$$\delta(x) = \begin{cases} \infty, & \text{if } x = 0 \\ 0, & \text{otherwise} \end{cases}$$

$$\int_{-\infty}^{\infty} \delta(x) dx = 1$$



$$p(x) = \delta(x - \mu) \quad !$$

Dirac Distribution



- **Dirac function / Dirac impulse**
 - Infinitely high value in an infinitely small range
 - Interesting fact: area under Dirac function (integral) = 1, hence
- The **Dirac distribution** is given by
 - Only one value μ is possible and has a likelihood of ∞
- **Multiple Dirac distributions** can be combined, for example
 - Three possible values: -1, 0, +1, with probabilities of 0.25, 0.5, 0.25
 - ➔ like a PMF over a discrete set of values, but in continuous form

$$\delta(x) = \begin{cases} \infty, & \text{if } x = 0 \\ 0, & \text{otherwise} \end{cases}$$

$$\int_{-\infty}^{\infty} \delta(x) dx = 1$$

$$p(x) = \delta(x - \mu) \quad !$$

$$p(x) = 0.25\delta(x + 1) + 0.5\delta(x) + 0.25\delta(x - 1)$$

Empirical Distribution

- **Goal:** describe probability distribution of measured data
- Given a set of **m measured values**
 - These were really **measured** → **high probability** assigned to all
 - All **other values** were not measured → **low (0) probability** assigned to all
- **Empirical distribution**
 - **Dirac** at each measured value
 - → the **true distribution** that was measured

$$x^{(1)}, \dots, x^{(m)}$$

$$\hat{p}(x) = \frac{1}{m} \sum_{i=1}^m \delta(x - x^{(i)})$$
 !

Overview of this lecture

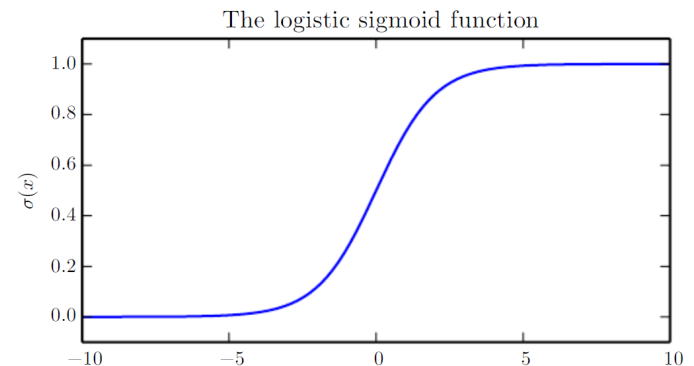
- Basics of probability
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Logistic Sigmoid Function

- **Definition** of logistic sigmoid
 - Note: $\sigma(x)$ is the **sigmoid function**, but σ is the **standard deviation** – these are entirely different!
- **Applications**
 - Logistic regression (ML)
 - Estimate parameter ϕ of a **Bernoulli distribution**
 - Estimate any output that must be between 0 and 1
- Note: **sigmoid saturates** for very high and very low inputs
 - Desirable → bounds output to $[0,1]$
 - But: will be problematic later on

$$\sigma(x) = \frac{1}{1 + \exp(-x)} \quad !$$

$$= \frac{\exp(x)}{1 + \exp(x)}$$



DL, Fig. 3.3

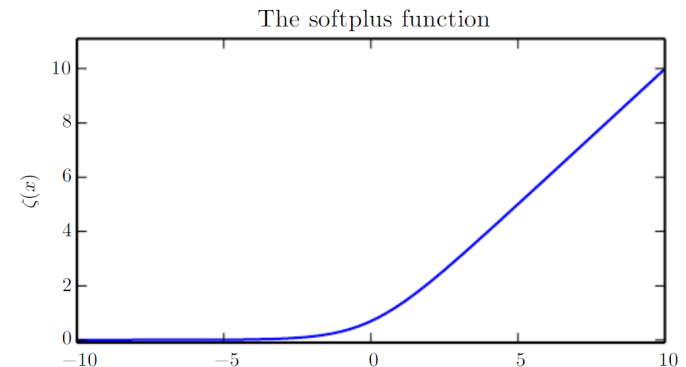
Softplus Function

- Plus function x^+
 - Cuts away the negative part
 - Also called the “**Rectified Linear**” function
- Softplus function $\zeta(x)$
 - “Soft” version of the plus function
 - No sharp edge at 0, but a **smooth transition**
 - Range of output: $(0, \infty)$
- Applications
 - Estimating the parameters β or σ of a **Gaussian distribution**
 - Estimate any output that must be **positive**

$$x^+ = \max(0, x)$$

$$\zeta(x) = \log(1 + \exp(x))$$

!



DL, Fig. 3.4

Properties of Sigmoid and Softplus

- Useful algebraic relationships
 - The sigmoid can also be written as shown on the right ($\exp(0)=1$)
 - Furthermore, its derivative can be expressed by the function itself
 - Not that surprising, since the sigmoid consists of exponential functions which are their own derivatives
 - The sigmoid has an interesting symmetry and the log of the sigmoid results in the flipped negative softplus
 - The derivative of the softplus turns out to be the sigmoid, i.e., the integral of the sigmoid results in the softplus

$$1 - \sigma(x) = \sigma(-x)$$

$$\zeta(x) - \zeta(-x) = x$$

$$\log \sigma(x) = -\zeta(-x)$$

$$\frac{d}{dx} \zeta(x) = \sigma(x)$$

$$\zeta(x) = \int_{-\infty}^x \sigma(y) dy$$

Properties of Sigmoid and Softplus

- Useful properties of the sigmoid and softplus

$$1 - \sigma(x) = \sigma(-x)$$

$$\zeta(x) - \zeta(-x) = x$$

- The sigmoid and softplus are closely related

$$\log \sigma(x) = -\zeta(-x)$$

$$\frac{d}{dx} \zeta(x) = \sigma(x)$$

$$\zeta(x) = \int_{-\infty}^x \sigma(y) dy$$

Applying functions to random variables

- Given a random vector $\mathbf{x}$ and a **deterministic function** $g(\mathbf{x})$, then $\mathbf{y}$ will also be a random vector
- If the distribution $p_x(\mathbf{x})$ is known, **how can we calculate $p_y(\mathbf{y})$** ?
- Intuitive (but **wrong**) approach:

$$y = g(x) \quad !$$

$$x = g^{-1}(y)$$

$$p_x(\mathbf{x}) = p_x(g^{-1}(\mathbf{y})) \neq p_y(\mathbf{y})$$

– Example:

$$x \sim U(0, 1)$$

$$y = f(x) = \frac{x}{2}$$

– The wrong approach would lead to:

$$p_y(y) = p_x(2y)$$

– Integrating the pdf results in:

$$\int p_y(y) dy = \frac{1}{2}$$

– Reason: rescaling required!

Applying functions to random variables

- **Rescaling** the probability density functions **correctly**

- **Example:** trivial, multiply $p_y(y)$ by 2
- In general: **equation on the right** must hold for any **infinitesimally small region** dx and dy
- **→ integral** over small regions of p_x and p_y **must be equal**

$$|p_y(g(x))dy| = |p_x(x)dx|$$

- **Solving** for $p_x(x)$ and $p_y(y)$ results in:

$$p_y(y) = p_x(g^{-1}(y)) \left| \frac{\partial x}{\partial y} \right|$$

$$p_x(x) = p_y(g(x)) \left| \frac{\partial g(x)}{\partial x} \right|$$

- Example $x = 2y$
 $\left| \frac{dx}{dy} \right| = 2$

- **→ multiply** $p_y(y)$ by 2

Applying functions to random variables

- **Generalization** to random vectors

$$p_x(\mathbf{x}) = p_y(g(\mathbf{x})) \left| \det \left(\frac{\partial g(\mathbf{x})}{\partial \mathbf{x}} \right) \right|$$

- Because $g(\mathbf{x}) = \mathbf{y}$, the term $\frac{\partial g(\mathbf{x})}{\partial \mathbf{x}}$ is a **matrix**

- This is called the **Jacobian matrix J**, which contains the **partial derivatives of all outputs** with respect to **all inputs**
- **Determinant: measure of volume** → measures, how much “space” is stretched or shrinked

$$\frac{\partial g(\mathbf{x})}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial g_1(\mathbf{x})}{\partial x_1} & \cdots & \frac{\partial g_1(\mathbf{x})}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial g_m(\mathbf{x})}{\partial x_1} & \cdots & \frac{\partial g_m(\mathbf{x})}{\partial x_n} \end{bmatrix}$$

- For large vectors $\mathbf{x}$ and $\mathbf{y}$, the Jacobian matrix can get huge!
 - Example: for $x=1000$, $y=1000$, the Jacobian contains *1 Mio* elements

Overview of this lecture

- Basics of probability
- Probability distributions
- Important functions
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Information Theory

- **Information theory** is a mathematical framework that **uses probability theory** to **describe information**
- Any **measure of information** should satisfy the **three rules** below

1. **Likely events** should have **low information** content.
2. **Less likely events** should have a **higher information** content.
3. **Independent events** should have **additive** information.

Self-Information

- Given an **event** x that occurs with **probability** $P(x)$

- The **self-information** $I(x)$ is defined as:

- That is, the **information content** of (one) **event** x happening
- Equivalently, this can be written as:

$$I(x) = \log\left(\frac{1}{P(x)}\right)$$

$$I(x) = -\log P(x) \quad !$$

- **Base** of the logarithm
 - Not important, corresponds to a change in unit
 - Base 2 → Unit is called “bits”
 - Base e → Unit is called “nats”

Shannon Entropy

- Self-information → single event
- **Shannon entropy $H(x)$ → average self-information over an entire distribution $P(x)$**
 - **Expected information content** when repeatedly drawing events from $P(x)$
 - → depends only on the distribution $P(x)$, hence often the notation $H(P)$ is used
- **Self-information** is the **number of bits** required for encoding one **single event**
- **Shannon entropy** is the **number of bits** required for encoding a **stream of events**

$$H(x) = \mathbb{E}_{x \sim P}[I(x)] = -\mathbb{E}_{x \sim P}[\log P(x)] !$$

$$H(P) = H(x)$$

Shannon Entropy

- Example

- Given a **binary random variable** (**Bernoulli** distribution) with the parameter

$$\phi = P(x = 1)$$

- If $\phi = 0$, x is always 0 $\rightarrow$ no information content
- If $\phi = 1$, x is always 1 $\rightarrow$ no information content

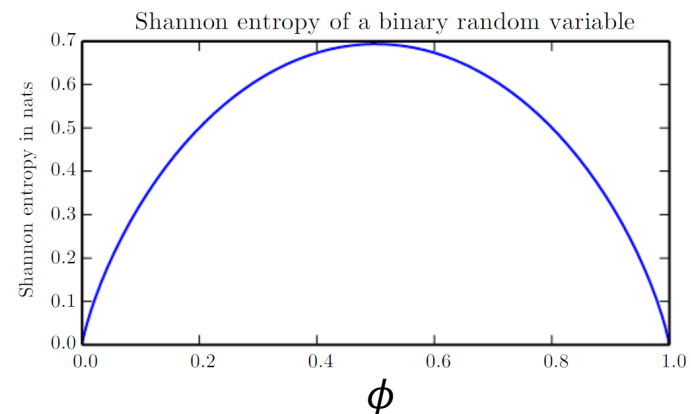
- If $\phi = 0.5$, then

$$I(x = 0) = \log\left(\frac{1}{0.5}\right) = \log(2)$$

$$I(x = 1) = \log\left(\frac{1}{0.5}\right) = \log(2)$$

$$H(x) = E[I(x)] = \log(2)$$

- $\rightarrow$ requires 1 bit or 0.7 nats



DL, Fig. 3.5

Kullback-Leibler (KL-) Divergence

- Measure **how different** two distributions **P** and **Q** are.
 - If $P(x) = Q(x)$, then:
 - The higher $D_{KL}(P \parallel Q)$, the more different P and Q are
- Note: the KL-divergence is **not symmetric**
 - Hence, it is **not a** mathematically valid **distance metric**

$$\begin{aligned} D_{KL}(P \parallel Q) &= \mathbb{E}_{x \sim P} \left[\log \frac{P(x)}{Q(x)} \right] \\ &= \mathbb{E}_{x \sim P} [\log P(x) - \log Q(x)] \end{aligned}$$

!

$$D_{KL}(P \parallel Q) = E_{x \sim P}[\log(1)] = 0$$

$$D_{KL}(P \parallel Q) \neq D_{KL}(Q \parallel P)$$

KL-Divergence and Cross-Entropy

- The **KL-divergence** can be decomposed into two parts:
 - The first term is the **Shannon entropy** $H(P)$
 - The second term is called the **cross-entropy** $H(P, Q)$ between the distributions P and Q
- The **cross-entropy** is defined as:

$$D_{\text{KL}}(P \| Q) = \mathbb{E}_{x \sim P} [\log P(x)] - \mathbb{E}_{x \sim P} \log Q(x)$$

$$D_{\text{KL}}(P \| Q) = -H(P) + H(P, Q)$$

$$H(P, Q) = -\mathbb{E}_{x \sim P} \log Q(x) \quad !$$

Information Theory

- **Meaning** of measures in information theory

Measure		Meaning
Shannon entropy	$H(P)$	Number of bits required to send symbols from P using an encoding optimally designed for P
Cross-entropy	$H(P, Q)$	Number of bits required to send symbols from P using an encoding optimally designed for Q
KL-divergence	$D_{\text{KL}}(P\ Q)$	Extra number of bits required, when sending symbols from P with an encoding optimally designed for Q instead of P

- Typically, in information theory, we assume:
 $0 \log(0) = 0$

Information Theory

How is this useful in Deep Learning?

- Given a set of m **training data** points
- The **empirical data distribution** is then:

$$\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(m)}$$
$$\hat{p}(\mathbf{x}) = \frac{1}{m} \sum_{i=1}^m \delta(\mathbf{x} - \mathbf{x}^{(i)})$$

- The **neural network predicts** a distribution
- **Training process:** make $p_{\text{model}}(\mathbf{x})$ as similar as possible to $\hat{p}(\mathbf{x})$

- This can be done by **minimizing the KL-divergence**
– Or equivalently: **minimize the cross-entropy**

$$p_{\text{model}}(\mathbf{x})$$

$$D_{\text{KL}}(P \| Q)$$

$$H(P, Q)$$

Overview of this lecture

- Basics of probability
- Probability distributions
- Important functions
- Information theory

Summary: Probability Theory

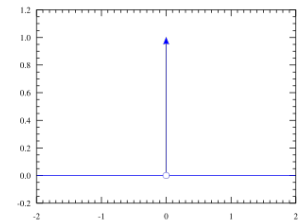
Type of rv	Values of rv	Description of rv with ...
Continuous	Real numbers	Probability density function (pdf), $p(x)$
Discrete	Discrete set	Probability mass function (pmf), $P(x)$

- For continuous rv $P(a \leq x \leq b) = \int_{[a,b]} p(x)dx$
- Bayes rule $P(x | y) = \frac{P(x)P(y | x)}{P(y)}$ where $P(y) = \sum_x P(y | x)P(x)$
- Expectation $\mathbb{E}_{x \sim P}[f(x)] = \sum_x P(x)f(x)$ or $\mathbb{E}_{x \sim p}[f(x)] = \int p(x)f(x)dx$
- Variance $\text{Var}(f(x)) = \mathbb{E}[(f(x) - \mathbb{E}[f(x)])^2]$
- Covariance $\text{Cov}(f(x), g(y)) = \mathbb{E}[(f(x) - \mathbb{E}[f(x)])(g(y) - \mathbb{E}[g(y)])]$
- Covariance matrix

$$\begin{matrix} & \begin{matrix} x & y & z \end{matrix} \\ \begin{matrix} x \\ y \\ z \end{matrix} & \begin{bmatrix} \text{var}(x) & \text{cov}(x, y) & \text{cov}(x, z) \\ \text{cov}(x, y) & \text{var}(y) & \text{cov}(y, z) \\ \text{cov}(x, z) & \text{cov}(y, z) & \text{var}(z) \end{bmatrix} \end{matrix}$$

Summary: Probability Distributions

- Uniform distribution $P(x = x_i) = \frac{1}{k}$ or $u(x; a, b) = \frac{1}{b-a}$ $x \in [a, b]$
- Bernoulli distribution $P(x = 1) = \phi$ and $P(x = 0) = 1 - \phi$
 - Multinoulli: over multiple states
- Gaussian distribution $\mathcal{N}(x; \mu, \sigma^2) = \sqrt{\frac{1}{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x - \mu)^2\right)$
 $\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \sqrt{\frac{1}{(2\pi)^n \det(\boldsymbol{\Sigma})}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$
- Exponential distribution $p(x; \lambda) = \lambda \mathbf{1}_{x \geq 0} \exp(-\lambda x)$
- Laplace distribution $\text{Laplace}(x; \mu, \gamma) = \frac{1}{2\gamma} \exp\left(-\frac{|x - \mu|}{\gamma}\right)$
- Dirac distribution $p(x) = \delta(x - \mu)$
- Empirical data distribution $\hat{p}(x) = \frac{1}{m} \sum_{i=1}^m \delta(x - x^{(i)})$



Summary: Information Theory

- Self-information $I(x)$ of one event: $I(x) = -\log P(x)$
- Shannon Entropy $H(x)$ $H(x) = \mathbb{E}_{x \sim P}[I(x)] = -\mathbb{E}_{x \sim P}[\log P(x)]$
 - Average self-information over all outcomes of a random variable
 - Number of bits required to transmit symbols drawn from P with a code designed for P
- Cross-Entropy $H(P, Q)$ $H(P, Q) = -\mathbb{E}_{x \sim P} \log Q(x)$
 - Number of bits required to transmit symbols drawn from P with a code designed for Q
- Kullback-Leibler Divergence $D_{KL}(P \parallel Q)$ $D_{KL}(P \parallel Q) = \mathbb{E}_{x \sim P} [\log P(x) - \log Q(x)]$
 - Measures similarity between two distributions P and Q $= H(P, Q) - H(P)$
 - Note: $D_{KL}(P \parallel Q) \neq D_{KL}(Q \parallel P)$

Overview of DL-Lecture

